Construction Chemicals

SUBMITTAL DOCUMENT

Mixcoat PU

Two-Component Polyurethane Waterproofing & Protective Membrane



Crack Bridging
Elongation ≥ 200 %

UV Resistant
For exposed areas

Chloride Barrier
96 % reduction

Adhesion
2 N/mm²

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

Mixcoat PU — Two-Component Polyurethane Waterproofing & Protective Membrane

Project Name		**Document No.**	OMX-MS-MIXCOATPU
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Mixcoat PU	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Mixcoat PU on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers surface preparation, priming, detailing, mixing and application of Mixcoat PU, a two-component polyurethane membrane, for waterproofing and protection of roofs, terraces, floors, façades and ceilings, including its use as top coat in the O MIX PU Protective system.

- Inspection and acceptance of the substrate and the work area.
- Surface preparation and repair of defects.
- Priming of the prepared surface.
- Mixing and application of the material as described in this document.
- Curing, protection and handover of the completed area.
- Quality control, inspection and documentation of each stage.

3. Definitions

Term	Meaning
Pot life	Time the mixed material remains workable after Part A and Part B are combined.
Recoat interval	Minimum and maximum time between coats to achieve inter-coat adhesion.
WFT / DFT	Wet film thickness measured during application / dry film thickness after cure.
Dew point	Temperature at which moisture condenses; the substrate must be at least 3 °C above it.
Kit / set	Pre-weighed Part A and Part B supplied to be mixed completely.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
Mixcoat PU	Polyurethane membrane, 20 kg and 5 kg sets (A + B)
Primer	O MIX PU primer or O-Prime SF; wet-surface primer on damp substrates
Repair mortar	For cracks, holes and coves
Reinforcement	Fibre mesh or geotextile for joints, corners and cracks
Quartz sand	0.3 – 0.8 mm, dry, for anti-slip finish where required

5.1 Key product data

Property	Typical Value
Mixing ratio (A : B)	1 : 1 by weight
Hardness	Shore A 80
Tensile strength	4.2 N/mm²
Elongation at break	≥ 200 %
Adhesion to concrete	2 N/mm²
Water vapour transmission	43 g/m²/day
Chloride ion reduction	96 %
Recoat interval	16 hours at 25 °C (max. 48 hours)

6. Tools & Equipment

- Grinder / shot blaster
- Industrial vacuum
- Slow-speed mixer
- Rollers
- Brushes

- Squeegee
- Notched trowel
- Spiked roller
- Wet-film gauge
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Wet-film thickness gauge
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Goggles
- Overalls
- Safety shoes

7. Storage & Handling

- 12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, protected from frost and direct sunlight.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep Part A and Part B of each kit together and do not mix parts from different batches.
- Store away from heat, sparks and ignition sources.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Check that the substrate is sound and at least 28 days old.
2. Remove laitance, oil, grease, curing compounds and loose material by grinding or shot blasting.
3. Open and repair cracks with repair mortar; fill holes and voids.

4. Form 45° coves at wall/floor junctions with repair mortar.
5. Moisten the surface where the wet-surface primer is used, leaving no standing water.
6. Vacuum all dust and apply the primer by roller; allow to become tack-free (6 – 8 hours).
7. Measure substrate temperature, air temperature and relative humidity; confirm the substrate is at least 3 °C above dew point.
8. Protect adjacent surfaces, drains and fixtures with masking tape and sheeting.
9. Obtain QA/QC approval of the prepared surface before starting the next step.

10. Mixing

1. Condition materials to between +15 °C and +25 °C before mixing.
2. Check the batch numbers and expiry dates of both parts and record them.
3. Stir Part A to a uniform consistency.
4. Add the complete Part B to Part A (1 : 1 by weight).
5. Mix 2 – 3 minutes with a slow-speed mixer, scraping sides and bottom, until uniform.
6. Mix only the quantity that can be applied within the pot life.

11. Application

1. Apply a first coat to joints, corners and cracks and embed the fibre mesh or geotextile while wet.
2. Apply the first full coat by squeegee or notched trowel and back-roll to an even film.
3. Apply the second coat crosswise within 48 hours (16 hours at 25 °C).
4. Build up to a total of 1.0 – 1.2 kg/m² per mm dry film in 2 – 3 coats.
5. Broadcast quartz sand into the last wet coat where an anti-slip finish is required.
6. Work in manageable bays, keeping a wet edge to avoid lap marks.
7. Check consumption by area and wet film thickness regularly during application and record the results.

12. Curing & Protection

- Allow each coat to cure before recoating, within the 48-hour window.
- Protect from rain, standing water and traffic until fully cured.
- Install protection layers (screed, tiles) only after the membrane has cured, where specified.
- Protect the applied area from water, dust, other trades and traffic until the stated cure times are reached.
- Barricade the area and display warning signs.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Substrate	Before priming	Sound, clean, ≥ 28 days, cracks repaired, coves formed	QA/QC
Climate	During work	+10 °C to +35 °C, RH ≤ 85 %, 3 °C above dew point	Site Engineer

Check	Stage	Acceptance Criteria	By
Primer	Before membrane	Uniform, tack-free	QA/QC
Membrane thickness	During application	1.0 – 1.2 kg/m² per mm; WFT checked	QA/QC
Continuity	After curing	No pinholes, blisters or thin spots; details reinforced	QA/QC
Water test	After curing (roofs)	Flood test 24 – 48 h, no leakage	QA/QC + Consultant

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point	Start of shift and every 4 hours	Daily climate log
Surface preparation	Visual, moisture, soundness	Each area	Inspection request (hold point)
Application	Consumption, thickness, appearance	Each coat / area	Daily progress report
Final inspection	Visual, continuity, finish	Each completed area	Handover checklist

14. Defects & Remedies

Defect	Likely Cause	Remedy
Pinholes / bubbles	Outgassing from porous substrate; air entrapped in mixing	Seal with primer; mix at low speed; de-aerate with spiked roller; sand and recoat
Blushing / tacky surface	High humidity or low temperature during cure	Wash and abrade; recoat in suitable conditions
Poor adhesion	Contaminated, weak or damp substrate; recoat window exceeded	Remove, re-prepare surface and reapply
Soft spots	Incorrect mixing or ratio	Remove affected area and reapply with full kits
Uneven colour / lap marks	Loss of wet edge; different batches	Abrade and apply a further coat from one batch

15. Health, Safety & Environment

Hazard	Control Measures
Skin and eye contact	Gloves, goggles, overalls; eye-wash available
Inhalation of vapours / dust	Ventilation; masks as required
Slips and trips	Housekeeping; barricades and signs
Manual handling	Correct lifting technique; mechanical aids
Electrical tools	Inspected tools; protected cables

- Wear chemical-resistant gloves, safety goggles and overalls.
- Avoid skin and eye contact; uncured material may cause sensitisation.
- Ensure good ventilation in enclosed areas.
- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean tools with water immediately after use; cured material can only be removed mechanically.

17. Limitations

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Do not apply in rain or when rain is expected before the material has set.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Substrate sound and clean	☐ Yes ☐ No	
Cracks repaired and coves formed	☐ Yes ☐ No	
Primer applied and tack-free	☐ Yes ☐ No	
Details reinforced with mesh	☐ Yes ☐ No	
Complete sets mixed at 1 : 1	☐ Yes ☐ No	
First coat applied uniformly	☐ Yes ☐ No	
Second coat within 48 h	☐ Yes ☐ No	
Total consumption verified	☐ Yes ☐ No	
Membrane continuous, no defects	☐ Yes ☐ No	
Flood test passed (roofs)	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

Mixcoat PU — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Protective Coating

Mixcoat PU

Two-Component Polyurethane
Waterproofing & Protective Membrane



Product Description

Mixcoat PU is a two-component, elastic polyurethane membrane for waterproofing and protection of roofs, terraces, floors, façades and ceilings. It forms a seamless, UV-resistant film that bridges cracks, protects concrete against carbonation and chloride ingress, and can be reinforced with fibre mesh at joints and details.

Uses

- Waterproofing of roofs, terraces and balconies.
- Protective membrane for floors, façades and ceilings.
- Top coat in the O MIX PU Protective system over epoxy body coats.
- Protection of concrete against carbonation and chlorides.
- Waterproofing of wet areas, with optional fibre-mesh reinforcement.

Advantages

- Seamless, elastic and crack-bridging.
- UV resistant for exposed areas.
- Elongation ≥ 200 %.
- Reduces chloride ingress by 96 %.
- Water cleaning of tools.

Application

1. **Surface Preparation**
   Clean, sound substrate; repair cracks; form coves at corners and prime.
2. **Mixing**
   Add Part B to Part A (1 : 1 by weight) and mix 2 – 3 min with a slow-speed mixer.
3. **Application**
   Apply by squeegee or notched trowel in at least two coats, 1.0 – 1.2 kg/m² per mm.

Consumption

1.0 – 1.2 kg/m²
per 1 mm dry film, in 2 – 3 coats
0.5 kg/m² when used as top coat in the PU Protective system.

Packaging

20 kg sets (Part A 10 kg + Part B 10 kg) and 5 kg sets (2.5 kg + 2.5 kg).

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, protected from frost and direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	1 : 1 by weight	—
Hardness	Shore A 80	—
Tensile strength	4.2 N/mm²	—
Elongation at break	≥ 200 %	—
Adhesion to concrete	2 N/mm²	—
Water vapour transmission	43 g/m²/day	—
Chloride ion reduction	96 %	—
Recoat interval	16 hours at 25 °C (max. 48 hours)	—
Primer tack-free time	6 – 8 hours	—

Consumption

Application	Coats	Consumption
Waterproofing membrane, 1 mm	2 – 3	1.0 – 1.2 kg/m²
Top coat in PU Protective system	1 – 2	0.5 kg/m²
Reinforced detail (joints, corners)	2 + mesh	as required

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Substrate clean and sound; damp but without standing water when the wet-surface primer is used.
- Concrete at least 28 days old.

Application Notes

- Prime with O MIX PU or solvent-free epoxy primer; use the wet-surface primer on damp substrates.
- Apply the membrane after the primer is tack-free (6 – 8 hours).
- Apply at least two coats; apply the next coat within 48 hours (16 hours at 25 °C).
- Reinforce joints, corners and cracks with fibre mesh or geotextile embedded in the first coat.
- Broadcast quartz sand into the final wet coat where an anti-slip finish is required.

Health & Safety

- Wear chemical-resistant gloves, safety goggles and overalls.
- Avoid skin and eye contact; uncured material may cause sensitisation.
- Ensure good ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools with water immediately after use; cured material can only be removed mechanically.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

Mixcoat PU — Submittal Document

To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

Mixcoat PU — Submittal Document

To be attached: approvals and certificates, where available.